Summary in Graph

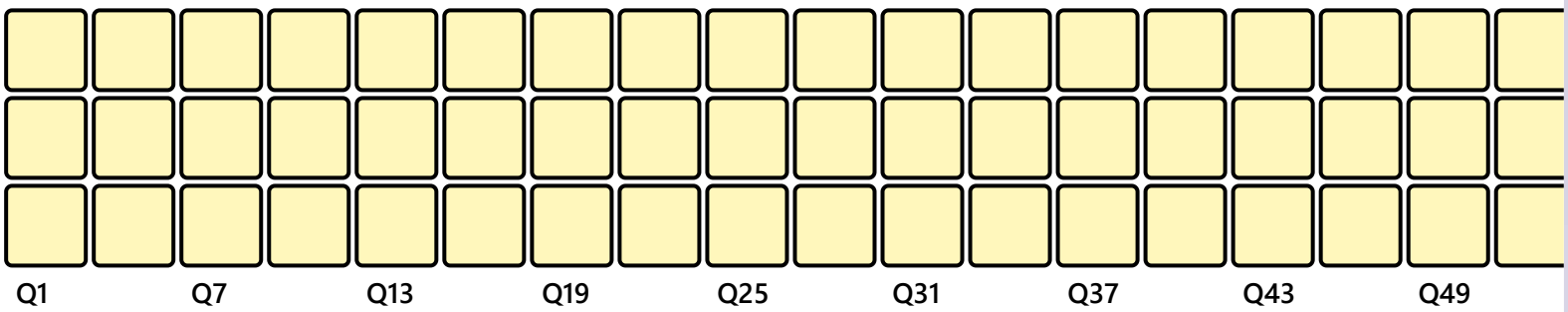
Exam Summary (GO Classes CS 2026 | All India Mock Test 2)

Qs. Attempted:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)
Correct Marks:	0 (0 + 0)
Penalty Marks:	0 (0 + 0)
Resultant Marks:	0 (0 + 0)

Total Questions:	65 (30 + 35)
Total Marks:	100 (30 + 70)
Exam Duration:	180 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (85) out of 846 users

EXAM RESPONSE EXAM STATS FEEDBACK

Aptitude

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Analytical Aptitude

A university schedules several exams on a single day.
Each exam is defined by a start time and an end time, and once an exam begins, it occupies one classroom continuously for its entire duration.
Two exams cannot be conducted in the same classroom if their time intervals overlap, including the case where one exam starts exactly at the same time another exam ends.

Given this scheduling constraint, determine the minimum number of classrooms required so that all exams can be conducted without any conflict.
Which of the following best describes this minimum number?

A. The total number of exams scheduled during the day

B. The maximum number of exams that overlap at any single point in time

C. The minimum number of exams that overlap at any single point in time

D. The number of distinct start times among all exams

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

Which of the following could be an integer?

A. The average of two consecutive integers

B. The average of three consecutive integers

C. The average of four consecutive integers

D. The average of six consecutive integers

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

Katrina has a wheat business. She purchases wheat from a local wholesaler at a particular cost per pound. The price of the wheat at her stores is $\$3$ per pound. Her faulty spring balance reads 0.9 pounds for a pound. Also, in the festival season, she gives a 10% discount on the wheat. She found that she made neither a profit nor a loss in the festival season. At what price did Katrina purchase the wheat from the wholesaler?

A. 2.43

B. 2.5

C. 2.7

D. 3

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 06sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

A, B and C are three distinct digits. If they are added as below,

$$\begin{array}{rrr} A & B & C \\ +A & B & C \\ +A & B & C \\ \hline C & C & C \end{array}$$

find out the value of A, B and C

A. $A = 3,\ B = 4,\ C = 5$

B. $A = 2,\ B = 3,\ C = 1$

C. $A = 5,\ B = 1,\ C = 3$

D. $A = 1,\ B = 8,\ C = 5$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Verbal Aptitude

Since she believed him to be both candid and trustworthy, she refused to consider the possibility that his statement had been _____

A. irrelevant

B. facetious

C. mistaken

D. insincere

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

A king ordered to make a crown from 8 kg of gold and 2 kg of silver. The goldsmith took away some amount of gold and replaced it by an equal amount of silver and the crown when made, weighed 10 kg. The king knows that underwater gold loses $\frac{1}{20}$th of its weight, while silver lost $\frac{1}{10}$th. When the crown was weighed underwater, it was 9.25 kg. How much gold was stolen by the goldsmith?

A. 1 Kg

B. 2 Kg

C. 3 Kg

D. 4 Kg

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Analytical Aptitude

One of four suspects A, B, C and D has committed a crime. A and D are always truthful, and B and C are always untruthful. C and D are identical twins and the interrogator does not know who is who. If A says, "D is innocent", B says, "A is guilty" and among C and D one says, "A is innocent" and the other says, "B is guilty", then which of the following is FALSE?

A. D said "A is innocent"

B. D is innocent

C. B is innocent

D. C is innocent

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

Two varieties A and B of rice cost Rs. 30 and Rs. 90 per kg, whereas two varieties C and D of pulses, Rs. 100 and Rs. 120 per kg, respectively. If at least one kg each of A and B and at least half a kg each of C and D have to be purchased, then the minimum and maximum costs of a total of 5 kg of these provisions are, respectively,

A. Rs. 150 and Rs. 600

B. Rs. 260 and Rs. 530

C. Rs. 290 and Rs. 470

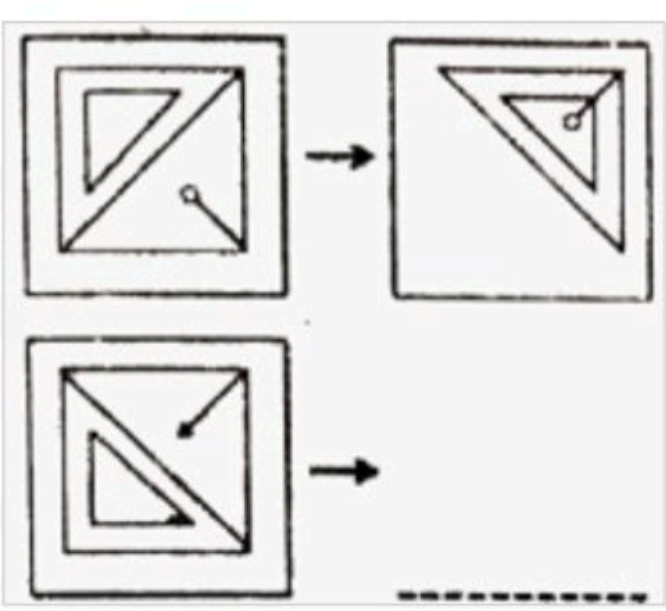
D. Rs. 370 and Rs. 460

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

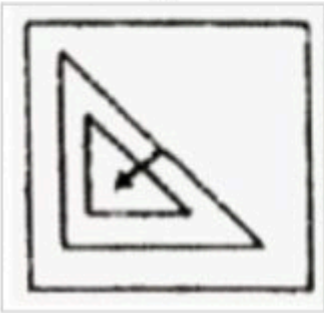
Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Spatial Aptitude

Observe the figures shown below. In the first pair of figures, the figure on the left undergoes a certain transformation to obtain the figure on the right. The transformation involves a systematic change in the outer boundary, the triangular structure inside the square, and the position of the arrow/circle element.

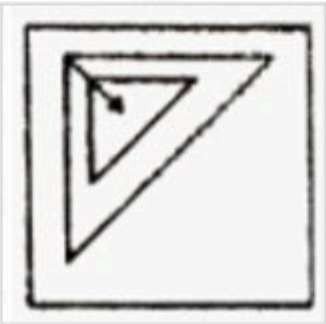
Apply the **same transformation rule** to the second figure given on the left and determine which option correctly represents the resulting figure that should appear in the blank space.



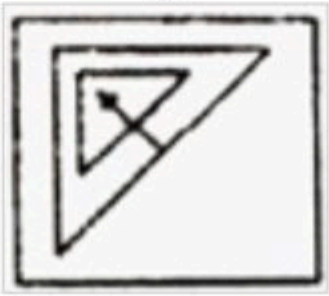
A.



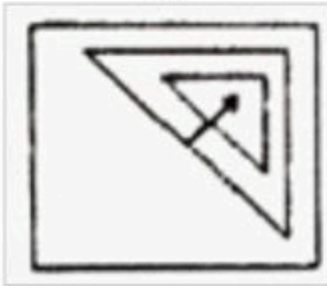
B.



C.



D.



Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

Three athletes, A, B, and C, competed pairwise in a d-meter race. Considering that each athlete had the same performance (i.e., the same speed) when competing against different opponents, and it is known that:

- A defeated B, finishing 20 meters ahead
- B defeated C, finishing 10 meters ahead
- A defeated C, finishing 28 meters ahead

We can then state that the race has:

A. 50 meters

B. 200 meters

C. 100 meters

D. 150 meters

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Technical

Q #1 | Multiple Select Type | Award: 1 | Penalty: 0 | Compiler Design

For each of the following grammars, $\{\mathrm{A}, \mathrm{B}, \mathrm{C}\}$ is the set of nonterminal symbols, A is the start symbol, $\{\mathrm{x}, \mathrm{y}, \mathrm{z}\}$ is the set of terminal symbols, and ε denotes the empty string.
Which of the following grammars is/are LL(1) ?

A.
$A ::= AxB \mid Cy$
$B ::= zAy$
$C ::= xA \mid \varepsilon$

B.
$A ::= Cz \mid xB$
$B ::= xC$
$C ::= yA \mid \varepsilon$

C.
$A ::= xA \mid By$
$B ::= zC \mid \varepsilon$
$C ::= zAx$

D.
$A \rightarrow xB \mid yAz$
$B \rightarrow xC$
$C \rightarrow y \mid \varepsilon$

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Numerical Type | Award: 1 | Penalty: 0 | Set Theory & Algebra

Let $A = \{1, 2, 3, 4\}$ and $R = \{(1, 2), (3, 4), (2, 4)\}$ be a relation on A. Let S be an equivalence relation on A such that $R \subset S$ and the number of elements in S is n. Then the minimum value of n is:

Your Answer: | Correct Answer: 16 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Select Type | Award: 1 | Penalty: 0 | Mathematical Logic

Select the **valid argument(s)**, where S_1, S_2 indicate the premises and S the conclusion:

A. S_1: If the horse is tired, then it will lose the race.
S_2: The horse was rested.
S: The horse won the race.

B. S_1: If the horse is tired, then it will lose the race.
S_2: The horse won the race.
S: The horse was rested.

C. S_1: If the horse is tired, then it will lose the race.
S_2: The horse lost the race.
S: The horse was tired.

D. S_1: If the horse is tired, then it will lose the race.
S_2: The horse was rested.
S: The horse lost the race.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Theory of Computation

The formal definition of a regular expression over alphabet Σ is:

1. The symbol $\varnothing$ is a regular expression.
2. The symbol ϵ is a regular expression.
3. For each $c \in \Sigma$, the symbol c is a regular expression.
4. If R_1 and R_2 are regular expressions, then the string ($R_1 \cup R_2$) is a regular expression.
5. If R_1 and R_2 are regular expressions, then the string ($R_1 R_2$) is a regular expression.
6. If R is a regular expression, then the string (R^*) is a regular expression.

For a regular expression e, let $L(e)$ be the language generated by e. If e is an expression that has no Kleene star * occurring in it, which of the following is true about e in general?

A. $L(e)$ is empty.

B. $L(e)$ is finite.

C. Complement of $L(e)$ is empty.

D. Both $L(e)$ and its complement are infinite.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Digital Logic

Considering A and B as two logical variables, the expression $(\text{not}(A)$ and $B)$ or (A and $\text{not}(B)$) is true:

A. for all values of A and B

B. whenever A is equal to B

C. whenever A is different from B

D. whenever A is false

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Digital Logic

Consider a logic circuit with three inputs, A, B, and C, and one output S. Considering that the output value S is equal to the input value A when inputs B and C are at the same logical level, and that the output value S is equal to the inverse of the input value A when inputs B and C are at opposite logical levels, select the alternative that presents the logical expression corresponding to the circuit.

A. $S = A \cdot (B \oplus \mathrm{C})$

B. $S = A \oplus \mathrm{B} \oplus \mathrm{C}$

C. $S = \bar{A} \oplus \mathrm{B} \oplus \mathrm{C}$

D. $S = A \oplus \overline{B \oplus \mathrm{C}}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Numerical Type | Award: 1 | Penalty: 0 | CO & Architecture

Consider a cache with 64-byte lines, 4 ways, and 4096 sets, for a machine with 40-bit physical addresses. Among all the physical blocks in the system, how many will be candidates for caching in a given set?

Your Answer: | Correct Answer: 4194304 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Numerical Type | Award: 1 | Penalty: 0 | CO & Architecture

Consider a processor with 128 registers and an instruction set of size 20. Each instruction has five distinct fields, namely, opcode, two source register identifiers, one destination register identifier, and a sixteen-bit immediate value. Each instruction must be stored in memory in a byte-aligned fashion. If a program has 250 instructions, the amount of memory (in bytes) consumed by the program text is _____

Your Answer: | Correct Answer: 1500 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Operating System

Consider a system with a total of $\mathbf{150}$ units of memory. Three processes are currently in the system, with their maximum memory needs and current allocations shown below:

Process	Max Demand	Currently Holds
$P1$	70	45
$P2$	60	40
$P3$	60	15

A new process $P4$ enters the system with a maximum memory requirement of 60 units.
Depending on its initial request, the state of the system may or may not remain safe.

Which of the following statements correctly describes the safety of the system?

A. If the system grants $P4$'s request of 25 units, the resulting state is unsafe because only 25 units remain and no process can finish.

B. If the system grants $P4$'s request of 25 units, the resulting state is safe because at least one of $P1$ or $P2$ can definitely complete.

C. If the system grants $P4$'s request of 35 units, the resulting state is safe because 15 units remain and $P3$ can finish first.

D. If the system grants $P4$'s request of 35 units, the resulting state is safe only if $P1$ finishes before $P2$ and $P3$.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Select Type | Award: 1 | Penalty: 0 | Operating System

Two processes compete for entering a critical section using a simple spin lock. Each process follows the pattern **entry** → **critical section** → **exit** exactly **once**. There is **no** `while(1)` or repetition of the critical section; each process tries to enter its critical section only **one time**.

It is possible that the system initially starts with only one of the processes in the ready state, while the other process becomes ready at some later time. (Both processes need not be ready at the start.)

```
// ENTRY SECTION
while (test_and_set(t));

// CRITICAL SECTION (executed once)

// EXIT SECTION
t = 0;
```

The instruction **test_and_set(X)** is atomic: it returns the old value of `X` and then sets `X = 1`.

Consider the following types of schedulers:

- **Priority scheduler:** Higher-priority processes always run before lower-priority ones.
- **Round-robin scheduler:** Preemptive, each process gets periodic CPU time.

Which of the following statements is/are correct regarding possible **deadlock**? (Multiple options may be correct.)

A. Under a priority scheduler, deadlock is possible.

B. Under a priority scheduler, deadlock cannot occur.

C. Under a round-robin scheduler, deadlock cannot occur.

D. Under a round-robin scheduler, deadlock is possible.

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Select Type | Award: 1 | Penalty: 0 | Computer Networks

A TCP sender is operating with a congestion window of 4 KB , and the receiver has advertised a window of 8 KB. At this moment, the sender has transmitted data up to byte number 20000. The receiver has acknowledged all bytes up to 16000.
Based on this information, identify which of the following byte numbers are guaranteed to have already been sent by the sender. **Assume 1K =** 10^3

A. 16200

B. 18000

C. 20001

D. 22000

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

When a host joins a network with no prior IP address, it must communicate with a DHCP server to obtain a valid IP configuration.
Which of the following options correctly represents the order of DHCP messages exchanged during this process?

A. DHCP Request → DHCP Discovery → DHCP Offer → DHCP ACK

B. DHCP Discovery → DHCP Offer → DHCP Request → DHCP ACK

C. DHCP Offer → DHCP ACK → DHCP Discovery → DHCP Request

D. DHCP Discovery → DHCP Request → DHCP Offer → DHCP ACK

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Select Type | Award: 1 | Penalty: 0 | Algorithms

I have just arrived by plane at Pittsburgh International Airport and I have forgotten where I have parked my car. There are n cars in the parking lot. I decide to look for my car with a simple serial search, examining, one by one, each and every car in the lot. This algorithm could be analyzed by considering its best case, average case, and worst case. Circle each of the following answers that are mathematically correct.

A. In the worst case, the algorithm is Big Omega(1).

B. In the best case, the algorithm is Big O(Log n).

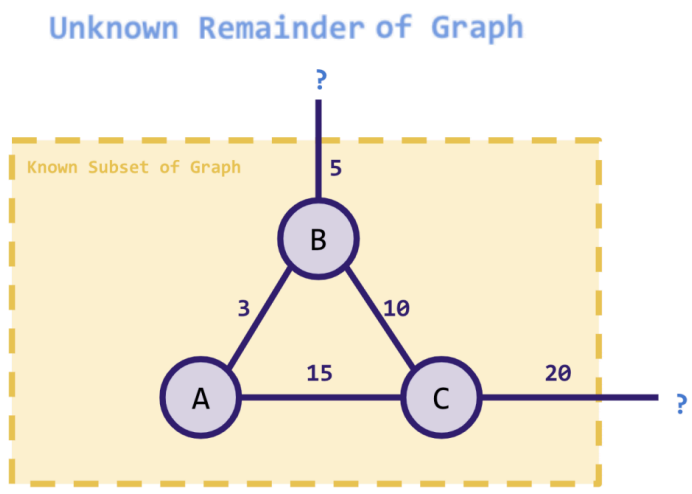
C. In the best case, the algorithm is Big O(n).

D. In the worst case, the algorithm is Big Theta(n^2).

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 Multiple Select Type | Award: 1 | Penalty: 0 | Algorithms

Consider the diagram below which shows only a partial snippet of a weighted undirected graph. There may be additional vertices and edges not shown.



Each edge marked with a $?$ connects to some unseen vertex. The two edges marked with a $?$ are the only edges that connect the shown portion to the rest of the graph. No other information about the remaining graph should be assumed.

For the edges shown in the snippet determine whether the following statements are **NEVER**, **ALWAYS**, or **SOMETIMES** true for a minimum spanning tree of the full graph. Here **SOMETIMES** means that the edge appears in a minimum spanning tree for some valid completions of the unseen part of the graph and does not appear for other valid completions.

Which of the following statements are correct? (Multiple options may be correct.)

A. The edge $A - B$ **ALWAYS** appears in a minimum spanning tree of the full graph.
B. The edge $B - ?$ **NEVER** appears in a minimum spanning tree of the full graph.
C. The edge $B - ?$ **SOMETIMES** appears in a minimum spanning tree of the full graph.
D. The edge $C - ?$ **SOMETIMES** appears in a minimum spanning tree of the full graph.

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Given positive integers $a_1, a_2, \ldots, a_k$.

For a positive integer x and $j \leq k$, let $D(x, j)$ denote the number of ways to write x as the sum of numbers chosen from $\{a_1, a_2, \ldots, a_j\}$, where each number can be used at most once and order does not matter.

For example, if $a_1 = 1, a_2 = 2, a_3 = 3$ and $a_4 = 4$, then there are exactly two ways to write $x = 5$ using $\{a_1, a_2, a_3, a_4\}$, namely:

$$5 = 2 + 3 = 1 + 4$$

Hence, $D(5, 4) = 2$.

For the purpose of this question, you do not need to worry about base cases.

Which of the following recurrence relations correctly computes $D(x, j)$?

A. $D(x, j) = D(x - a_j, j) + D(x, j - 1)$
B. $D(x, j) = D(x - a_j, j - 1) + D(x, j - 1)$

C. $D(x, j) = D(x - a_j, j - 1) + D(x - a_j, j)$
D. $D(x, j) = D(x, j - 1) - D(x - a_j, j - 1)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 Numerical Type | Award: 1 | Penalty: 0 | Data Structures

Consider the following max heap represented using an array. However, the array does not currently satisfy the max-heap property.

The array representation of the heap is given below (indexed from left to right starting at index 1):

$$100 \quad 75 \quad 90 \quad 70 \quad 80 \quad 72 \quad 76 \quad 15 \quad 9 \quad 8 \quad 14 \quad 13$$

Assume the standard max-heap property:

- For every node at index i, the value at index i is greater than or equal to the values of its children (if they exist).
- The left child of index i is at index $2i$, and the right child is at index $2i + 1$.

To transform the given array into a valid max heap, exactly how many swaps are required?

Your Answer: | Correct Answer: 1 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Suppose a hash table has m slots and only one of these slots contains a single element with key k, and the remaining slots are empty. Say we search in this table for x various other keys, none of which are equal to k. Assuming **simple uniform hashing**, what is the probability that at least one of these searches probes the slot containing k?

A. $\frac{x}{m}$

B. $1 - \left(1 - \frac{1}{m}\right)^x$

C. $\left(1 - \frac{1}{m}\right)^x$

D. $\frac{1}{m}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 Numerical Type | Award: 1 | Penalty: 0 | Operating System

Consider a system with four page frames and the following page access pattern:

$$A,\ B,\ A,\ C,\ D,\ E,\ F,\ A,\ B,\ C,\ F,\ A,\ E,\ F$$

The system uses the FIFO (First-In First-Out) page replacement policy. Initially, all frames are empty.

Determine the total number of page faults that occur for the given access pattern.

Your Answer: | Correct Answer: 11 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Suppose $f(x)$ is a function such that

$f'(x) = x - \frac{f(x)}{2}$.

If $f(1) = 2$, then evaluate $\lim_{x \to 1} \frac{f(x)-2}{(x-1)^2}$.

A. -1

B. $-1/2$

C. 0

D. $1/2$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C

Assuming "`s`" is a pointer to a struct, which of the following is equivalent to `(*s).data`?

A. `s->data`

B. `*s.data`

C. `*(s.data)`

D. `s.data`

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C

What will be the output of the given program?

```c
#include <stdio.h>

int main()
{
    unsigned int i;
    int sum = 0;

    for (i = 5; i >= 0; i--)
    {
        sum += i;
    }

    printf("%d", sum);
    return 0;
}
```

A. 5
B. 15
C. 1
D. None of the above

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C

Assume that x is a two's complement integer in the C programming language whose absolute value is small enough to avoid overflow. The operator ~ denotes the 1's complement, i.e., it flips every bit in the binary representation of x.

Which of the following expressions evaluates to $-x-1$?

A. $x+1$

B. $\sim x$

C. $\sim x+1$

D. $\sim(\sim x)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Suppose A is a 3×3 matrix such that

$$\det(A-\lambda I) = -\lambda^3 + 3\lambda^2 + \lambda - 3,$$

where I is the 3×3 identity matrix. Which of the following are TRUE of A ?

I. The trace of A is 3.
II. The determinate of A is -3.
III. The matrix A has eigenvalues -3 and 1.

A. I only
B. II only
C. III only
D. I and II only

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

If f is a continuously differentiable real-valued function defined on the open interval $(-1, 4)$ such that $f(3) = 5$ and $f'(x) \geq -1$ for all x, what is the greatest possible value of $f(0)$?

A. 3

B. 4

C. 5

D. 8

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Consider a tree with the following properties:

- Each internal node has exactly three children.
- The heights of the subtrees rooted at each child differ by at most 1.

Let $T(h)$ denote the minimum number of nodes in such a tree of height h.
Which of the following recurrence relations correctly describes $T(h)$?

A. $T(h) = 3T(h-1) + 1$

B. $T(h) = T(h-1) + T(h-2) + T(h-3) + 1$

C. $T(h) = T(h-1) + 2T(h-2) + 1$

D. $T(h) = 2T(h-1) + T(h-2) + 1$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Multiple Select Type | Award: 2 | Penalty: 0 | Compiler Design

If G is a context-free grammar with start symbol S, then G', the augmented grammar for G, is obtained by introducing a new start symbol S' and the production $S' \rightarrow S$.

In the context of LR(0) parsers, an LR(0) item (item for short) of grammar G' is a production of G' with a dot at some position of the body.

We divide the LR(0) items of G' into two classes:

1. *Kernel items:* The initial item $S' \rightarrow \cdot S$, and all items whose dots are not at the left end.
2. *Nonkernel items:* All items with their dots at the left end, except for $S' \rightarrow \cdot S$.

Each state of the LR(0) automaton represents a set of items. Which of the following is/are true about the states in LR(0) automaton for grammar G':

A. Each state is the closure of a set of kernel items.

B. Each state is the closure of a set of nonkernel items.

C. If all nonkernel items are removed from a state, they can always be regenerated by applying the closure operation to the remaining items.

D. If any kernel item is removed from a state, it can never be regenerated by applying the closure operation to the remaining items.

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Numerical Type | Award: 2 | Penalty: 0 | Compiler Design

The syntax-directed translation scheme given below translates a language with terminals a, b, c, and d into a language whose terminals are $1, 2, 3, 4, 5$, and 6. Using a bottom-up parser that executes the actions in parentheses immediately after reducing the corresponding rule, what is the output string produced by the translation for the input string "aaadbc"?

$$\begin{array}{ll} \mathrm{S} \rightarrow \mathrm{AS} & \{\text{print ``1''}\} \\ \mathrm{S} \rightarrow \mathrm{B} & \{\text{print ``2''}\} \\ \mathrm{A} \rightarrow \mathrm{a} & \{\text{print ``3''}\} \\ \mathrm{B} \rightarrow \mathrm{bC} & \{\text{print ``4''}\} \\ \mathrm{B} \rightarrow \mathrm{dB} & \{\text{print ``5''}\} \\ \mathrm{C} \rightarrow \mathrm{c} & \{\text{print ``6''}\} \end{array}$$

Your Answer: | Correct Answer: 3336452111 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Numerical Type | Award: 2 | Penalty: 0 | Combinatory

An entrance examination was prepared with 25 multiple-choice questions, each with 5 alternatives. In each question, at most one alternative can be selected, and leaving a question unanswered is also considered a possible way of answering. The number of candidates who took the exam was $63,127$. Consider the statement:

"At least two candidates answered the first k questions of the exam in an identical way."

What is the largest value of k for which we can guarantee that the statement is true?

Your Answer: | Correct Answer: 6 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Set Theory & Algebra

Let L be a lattice $[\mathrm{S}, \#]$, where S is the base set and $\#$ is a relation defined on S. If for every subset A of S, $[\mathrm{A}, \#]$ is also a lattice then L must be?

A. Total order

B. Bounded lattice

C. Not Distributive lattice

D. Boolean lattice

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Set Theory & Algebra

A 3-ary boolean function is a function that takes three boolean arguments and produces a boolean output. Let f and g be 3-ary boolean functions. We say that f and g are neighbours if f and g agree on at least one input and disagree on at least one input: that is, there exist $x,\ y,\ z$ such that $f(x,y,z) = g(x,y,z)$ and x', y', z' such that $f(x',y',z') \neq g(x',y',z')$.

Suppose we fix a 3-ary boolean function h. How many neighbours does h have?

A. 128

B. 132

C. 254

D. 256

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #31 | Numerical Type | Award: 2 | Penalty: 0 | Graph Theory

Consider a random graph G on 10 vertices where for each pair of vertices u, v, there is an edge (u, v) with probability $1/4$. What is the expected number of cycles of length 3 in this graph? (up to three places after the decimal point)

Your Answer: | Correct Answer: 1.875 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #32 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Let $\mathrm{R(A,B,C,D,E)}$ be a relational schema with the following functional dependencies:
$\mathrm{A} \rightarrow \mathrm{D},\ \mathrm{CD} \rightarrow \mathrm{E},\ \mathrm{E} \rightarrow \mathrm{D}$

The decomposition of R into $(\mathrm{A}, \mathrm{B}, \mathrm{C})$, $(\mathrm{B}, \mathrm{C}, \mathrm{D})$, and $(\mathrm{A}, \mathrm{C}, \mathrm{E})$

A. gives a lossless join, and is dependency preserving

B. gives a lossless join, but is not dependency preserving

C. does not give a lossless join, but is dependency preserving

D. does not give a lossless join and is not dependency preserving

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #33 | Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Two queries are equivalent on a database instance if they produce the same result relation when evaluated over that instance.

Consider a SQL database with two tables, $\mathrm{R}(\mathrm{A}, \mathrm{B})$ and $\mathrm{S}(\mathrm{C})$. Assume NULL values are not permitted in any attribute; make no other assumptions about the data.

Below are pairs of queries. In which of the following options, the two queries are equivalent on every possible instance of R and S ? (Select all correct options)

A.
```
Query1: Select A From R
Query2: Select A From R, S
```

B.
```
Query1: Select max(A) From R
Query2: Select A From R Where A >= all (select A from R)
```

C.
```
Query1: Select Distinct A From R
Query2: Select A From R Group By A
```

D.
```
Query1: Select Distinct A From R Group By A Having max(B) > 10
Query2: Select Distinct A from R where B > 10
```

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #34

Numerical Type | Award: 2 | Penalty: 0 | Databases

A B-tree of order M has the following properties:

- M denotes the max number of child nodes a node can point to.
- The max number of keys per node is $M-1$.
- The root node is either a leaf node or an internal node with 2 to M children and 1 to $M-1$ keys.
- The rest of the internal nodes have between $\left\lceil \frac{M}{2} \right\rceil$ and M children.
- All leaves are at the same depth, which makes the tree balanced.

What is the total number of key values stored in the internal nodes of a B-tree of order 5 with 1000 leaf nodes?

Your Answer: | Correct Answer: 999 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #35

Numerical Type | Award: 2 | Penalty: 0 | Databases

Let transactions $\mathrm{T1}$, $\mathrm{T2}$, and $\mathrm{T3}$ be defined to perform the following operations:

- $\mathrm{T1}$: Add one to A
- $\mathrm{T2}$: Double A

- $\mathrm{T3}$: Display the current value of A on the screen and then set A to one

(where A is some numeric item in the database).

Suppose the internal structure of $\mathrm{T1}$, $\mathrm{T2}$, and $\mathrm{T3}$ is as indicated below:

$\mathrm{T_1}$	$\mathrm{T_2}$	$\mathrm{T_3}$
$\mathrm{F_1}:$ fetch A into $\mathrm{t_1}$ $\mathrm{t_1} := \mathrm{t_1} + 1$	$\mathrm{F_2}:$ fetch A into $\mathrm{t_2}$ $\mathrm{t_2} := \mathrm{t_2} * 2$	$\mathrm{F_3}:$ fetch A into $\mathrm{t_3}$ display $\mathrm{t_3}$
$\mathrm{U_1}:$ update A from $\mathrm{t_1}$	$\mathrm{U_2}:$ update A from $\mathrm{t_2}$	$\mathrm{U_3}:$ update A from 1

Suppose transactions $\mathrm{T1}$, $\mathrm{T2}$, and $\mathrm{T3}$ are allowed to execute concurrently. If A has initial value zero, how many possible values are displayed by $\mathrm{T3}$?

Your Answer: | Correct Answer: 3 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #36

Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

In the following, assume that i, j are positive integers and that n is a fixed positive integer.

Which of the languages given below, is/are regular?

A. $\{a^i b^j \mid j = n - i\}$

B. $\{a^i b^j \mid j = n + i\}$

C. $\{a^i b^j \mid j = n * i\}$

D. $\{a^i b^j \mid j = n/i\}$ (”/” interpreted as integer division)

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #37

Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Let $\Sigma = \{a, b\}$. For a word $w \in \Sigma^*$, let $n_a(x)$ denote the number of a's in w and let $n_b(x)$ denote the number of b's in w. Consider the following language:

$$L := \{xy \mid x, y \in \Sigma^*, n_a(x) = n_b(y)\}$$

What can we say about L ? (Select all that apply)

A. L is regular

B. L is context-free

C. L is Σ^*

D. L is not Σ^*

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #38

Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

Let M be a Turing machine over the alphabet Σ. We denote by $\mathrm{ACCEPT}(M)$ the set of strings accepted by M. A language $L \subseteq \Sigma^*$ is called **Turing-recognizable** if there exists a Turing machine M such that $L = \mathrm{ACCEPT}(M)$. We use $\mathrm{TR}(L)$ to denote that the language L is Turing-recognizable. In this sense, analyze the following statements about two languages L_1 and L_2 over the alphabet Σ :

I. If $\mathrm{TR}(L_1)$ and $\mathrm{TR}(L_2)$, then $\mathrm{TR}(L_1 \cup L_2)$.

II. If $\mathrm{TR}(L_1)$, then $\mathrm{TR}(\Sigma^* \backslash L_1)$.

III. If $\mathrm{TR}(L_1)$ and $\mathrm{TR}(L_2)$, then $\mathrm{TR}(L_1 \cap L_2)$.

Which is correct?

A. Only I.

B. Only I and III.

C. I, II and III.

D. None

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #39 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

Let $M_0, M_1, M_2, \ldots$, be an effective enumeration of all Turing machines. Which of the following problems is (are) decidable?

I. Given a natural number n, does M_n starting with an empty tape halt in fewer than n steps?

II. Given a natural number n, does M_n starting with an empty tape halt in exactly n steps?

III. Given a natural number n, does M_n starting with an empty tape halt after at least n steps?

A. I only

B. I, II only

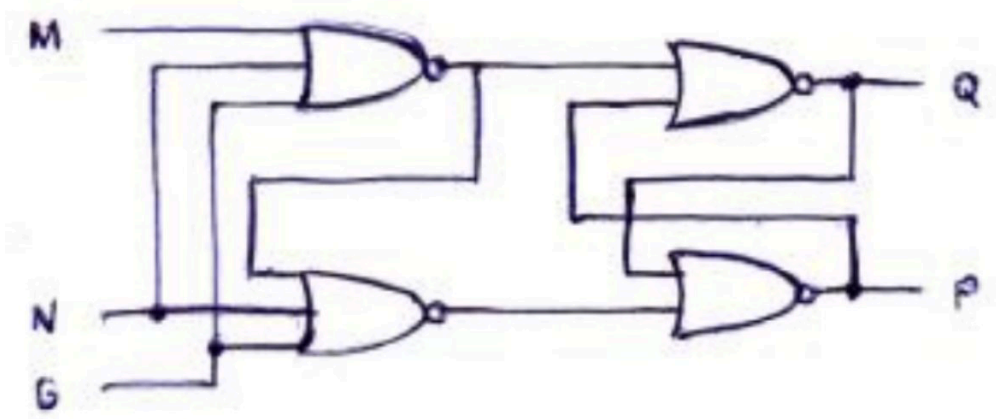
C. All

D. None

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #40 | Multiple Select Type | Award: 2 | Penalty: 0 | Digital Logic

Consider a sequential circuit shown below.



For which input values of M, N, and G, outputs P and Q hold their previous values? (Select all correct options)

A. $\mathrm{M}=1$ and G, N can be anything

B. $\mathrm{N}=1$ and G, M can be anything

C. $\mathrm{G}=1$ and M, N can be anything

D. $\mathrm{M}=1, \mathrm{G}=0,\ \mathrm{N}=0$

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #41

Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

Let $A=\langle a_1 a_0\rangle$ and $B=\langle b_1 b_0\rangle$ be two 2-bit numbers and $C=\langle c_3 c_2 c_1 c_0\rangle$ be a 4 -bit number, where $c_0=a_0, c_1=a_1 \oplus b_0$, $c_2=b_1 \oplus a_1 b_0$, and $c_3=a_1 b_0 b_1$ (' $\oplus$ ' denotes the XOR operation). If $X+Y$ and XY denote the arithmetic sum and product (of X and Y) respectively, which of the following is true?

A. $C=A+B$

B. $C=AB$

C. $C=2A+B$

D. $C=A+2B$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #42

Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

According to the IEEE standard, a 32-bit, single-precision, floating-point number N is defined to be

$$\mathrm{N}=(-1)^{\mathrm{S}} \times 1.\mathrm{F} \times 2^{\mathrm{E}-127}$$

where S is the sign bit, F the fractional mantissa, and E the biased exponent.

A floating-point number is stored as $\mathrm{S}:\mathrm{E}:\mathrm{F}$, where S, E, and F are stored in 1 bit, 8 bits, and 23 bits, respectively.

What is the decimal value of the floating-point number C1E00000 (hexadecimal notation)?

A. -12

B. -15

C. -26

D. -28

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #43 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

In the context of a single-level cache memory, suppose memory addresses A and B have different tags but the same index (for those caches that have tags and indices, that is). Assume the cache is empty prior to the accesses listed in the question.

In the access pattern "read A, write A, read B, write B, read A, write A", to minimize writes to the main memory we should pick which of the following?

A. a write-through fully-associative cache

B. a write-back fully-associative cache

C. a write-through direct-mapped cache

D. a write-back direct-mapped cache

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #44 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

Consider a CPU using a $5-$stage pipeline structure (IF, ID, EX, MEM, WB) with separate data and instruction memories, no data forwarding mechanism, writing to the register bank on the rising edge of the clock and reading on the falling edge of the clock, and the following set of instructions:

```
   I1: lw   $2, 100($5)
   I2: add  $1, $2, $3
   I3: sub  $3, $2, $1
   I4: sw   $2, 50($1)
5. I5: add  $2, $3, $3
   I6: sub  $2, $2, $4
```

Each stage takes exactly one clock cycle to complete. How many clock cycles are required for the execution of this code?

A. 10

B. 11

C. 16

D. 17

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #45 Multiple Select Type | Award: 2 | Penalty: 0 | Operating System

An operating systems student is attempting reverse engineering of a machine whose page table was lost after a crash. Only a small log of recent address translations is available, and the student wants to reconstruct the missing page table entries from this behavior.

The available address translation log is shown below.

Virtual address		Physical address
0x1063	$\rightarrow$	0x2063
0x584a	$\rightarrow$	0xe84a
0x4dfe	$\rightarrow$	Invalid
0x2a9f	$\rightarrow$	0x9a9f

The system configuration is as follows.

$$\text{Virtual address space} = 32 \text{ KB}, \quad \text{Physical memory} = 64 \text{ KB}, \quad \text{Page size} = 4 \text{ KB}$$

A simple linear page table is used, where each page table entry stores a valid bit and a physical frame number (PFN) and is written in the form $(\text{Valid bit}, \text{PFN})$. Let PTE denote the page table implemented as an array, where $\text{PTE}[i]$ represents the entry corresponding to virtual page number i.

Based only on the above log, select all options that must be correct.

A. $\text{PTE}[1] = (1,\ 2)$

B. $\text{PTE}[2] = (1,\ 0\text{x}9)$

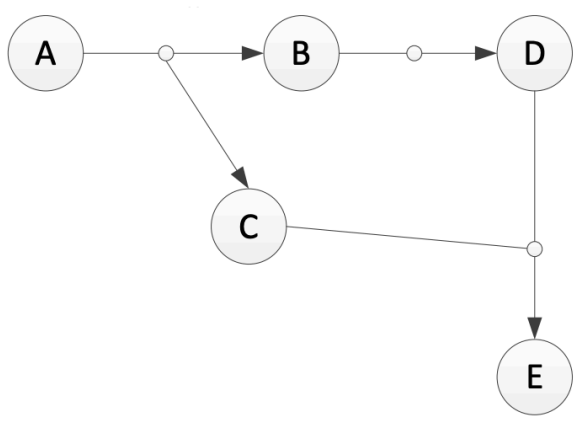
C. $\text{PTE}[4] = (0,\ _)$

D. $\text{PTE}[5] = (1,\ 0\text{xE})$

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #46 Numerical Type | Award: 2 | Penalty: 0 | Operating System

Consider the following process-ordering diagram:



The ordering constraints are:

- A must finish before B starts
- A must finish before C starts
- B must finish before D starts
- C and D must both finish before E starts

Each dependency of the form "X must finish before Y starts" is enforced using a semaphore.

What is the minimum number of counting semaphores required to correctly enforce all the above ordering constraints?

Your Answer: | Correct Answer: 3 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #47 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Computer Networks

A sender S communicates with a receiver D using a TCP variant where duplicate ACKs are ignored and loss detection occurs only via timeout. The retransmission timeout is $\mathrm{RTO} = 3$ RTTs. Transmission starts at $t = 0$ in slow start, packet numbering begins at 1, and the 14th packet is lost during its first transmission.

We assume continuous transmission at the sender: as soon as transmission of one window completes, transmission of the next window begins; i.e., the sender always has data to send.

The slow-start threshold is $\mathrm{ssthresh} = 18$ MSS.

Packets sent at the start of each RTT are partially shown below:

RTT start	Packets sent
1	1
2	2, 3
3	(Not given)
4	(Not given; packet 14 is lost)
5	?

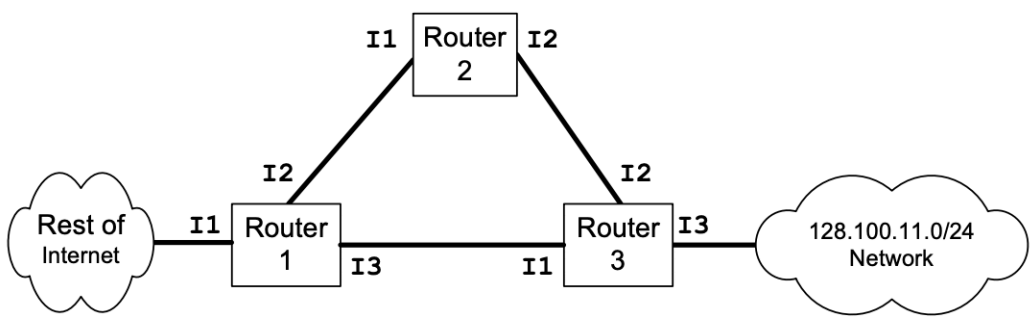
Which of the following correctly represents the packets transmitted at the start of RTT 5?

A. $16 – 27$
B. $16 – 29$
C. $14 – 27$
D. None of the above

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #48 | Multiple Select Type | Award: 2 | Penalty: 0 | Computer Networks

Consider the statically routed IP network consisting of three routers as shown in the topology below. Router 3 is connected to an Ethernet network $128.100.11.0/24$. The interfaces of each router are labeled as $I1, I2, I3$. All routers use longest prefix matching.



The routing tables configured on the three routers are shown below.

Router 1

Destination	Interface
$128.100.11.32/27$	$I2$
$128.100.11.0/24$	$I3$
$0.0.0.0/0$	$I1$

Router 2

Destination	Interface
$128.100.11.32/27$	$I2$
$0.0.0.0/0$	$I1$

Router 3

Destination	Interfac
$128.100.11.16/28$	$I2$
$128.100.11.0/24$	$I3$
$0.0.0.0/0$	$I1$

Due to the above routing configuration, packets destined to some IP addresses enter a routing loop.

Which of the following destination IP addresses, when received at Router 1, will result in packets looping indefinitely in the network?

A. 128.100.11.10

B. 128.100.11.18

C. 128.100.11.25

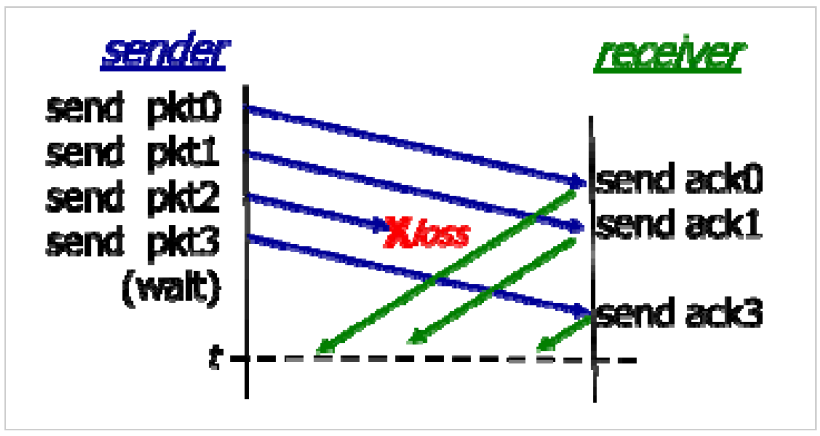
D. 128.100.11.40

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #49 | Multiple Select Type | Award: 2 | Penalty: 0 | Computer Networks

Consider the sliding window packet transmission scenario shown below.

- Packet 2 is lost during transmission
- At time t (indicated by the horizontal dashed line in the figure), the following ACKs are currently propagating from receiver to sender:
  - ACK0
  - ACK1
  - ACK3
- No ACK has yet been received by the sender by time t



Answer the following based only on future events at the sender after time t.

Which of the following statements are correct? (Select all that apply.)

A. If ACK0 is the next ACK received, the sender will advance its window and transmit packet 4.

B. If ACK1 is the next ACK received (and ACK0 is lost), the sender will advance its window to start at packet 2, but will not transmit any new packet.

C. If ACK2 is the next ACK received (and ACK0 and ACK1 are lost), the sender will advance its window and transmit packet 6.

D. If ACK3 is the next ACK received (and ACK0 and ACK1 are lost), the sender will mark packet 3 as acknowledged but will not advance the window.

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #50 | Numerical Type | Award: 2 | Penalty: 0 | Algorithms

Given two *sorted* arrays $A[1..m]$ and $B[1..n]$, we want to find the kth smallest element in the concatenation of the two arrays. Assume $k < m$ and $k < n$.

We are allowed unit-time random access to any indexed element of both lists.
Assume that all $m + n$ numbers are distinct.

If the worst-case running time of the best algorithm is expressed as

$$\Theta\left(n^a \log^b m + m^c \log^d n\right),$$

then the value of

$$a + 10b + 100c + 1000d$$

is _____.

Your Answer: | Correct Answer: 1010 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #51 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

The following shows a series of steps in order at certain points in a sorting algorithm, although there may be additional steps omitted in between. Which sorting algorithm(s) could be being run here? Select all that apply.

Initial Array:

```
10, 5, 8, 3, 4, 2, 9, 1, 6, 7
```

Some Later Step:

```
5, 6, 7, 10, 9, 8, 4, 3, 2, 1
```

Final Step:

```
1, 2, 3, 4, 5, 6, 7, 8, 9, 10
```

A. Insertion Sort

B. Selection Sort

C. In-place Heap Sort

D. Merge Sort

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #52 | Numerical Type | Award: 2 | Penalty: 0 | Algorithms

A Depth First Search (DFS) is performed on a graph with vertices $\{A, B, C, D, E, F, G, H, I, J, K, L\}$.

The vertices listed in increasing order of their discovery times are:

$A, B, C, D, E, F, G, H, I, J, K, L$

The vertices listed in increasing order of their finish times are:

$C, D, E, B, H, I, G, F, A, K, L, J$

Based on the above DFS information, the number of connected components in the graph is _____.

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #53 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

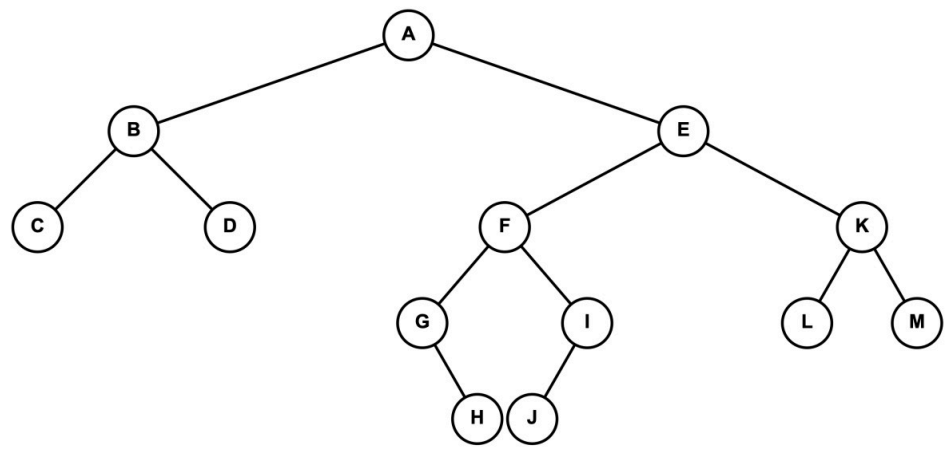
Consider the following definitions of path orders of a binary tree:

Order A: Order B:

```
if the binary tree is not empty:
{
  visit the root;
  traverse the left subtree in B-order;
  traverse the right subtree in
    B-order;
}
```

```
if the binary tree is not empty:
{
  visit the root;
  traverse the right subtree in
    Order A;
  traverse the left subtree in
    Order A;
}
```

Consider the following binary tree. The path of the binary tree shown is:



If the root of the given binary tree is passed to Order A, what will be the resulting sequence of node visits?

A. $\text{A B C D E K L M F I J G H}$
B. $\text{A B C D E F G H I J K L M}$
C. $\text{A B D C E K L M F G H I J}$
D. $\text{A B E C D F K G I L M H J}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #54 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in C

Consider the following C program:

```
#include <stdio.h>

int mystery(int *p, int n)
{
    if (n <= 1)
        return *p;

    int y = mystery(p + 1, n - 1);

    return (*p > y) ? *p : y;
}

int main()
{
    int a[5] = {12, 7, 45, 23, 9};

    printf("%d", mystery(a, 5));
    return 0;
}
```

What will be the output of the program?

A. 12
B. 23

C. 45
D. 9

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #55 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in C

Consider the following C program:

```c
#include <stdio.h>

int main()
{
    int a[4] = {5, 6, 7, 8};

    int *p = a + 3;

    printf("%d", p[-2]);

    return 0;
}
```

What will be the output of the above program?

A. 5
B. 6
C. 7
D. Undefined behavior

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

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